

Apala Pramanik

Lincoln, Nebraska, USA

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Personal Profile

PhD researcher at the University of Nebraska–Lincoln specializing in ML-driven beam management and power control for mmWave 5G/6G vehicular networks (V2I/V2X). Experienced in SDR-based hardware-in-the-loop testbeds, ray-tracing channel modeling (beamforming, channel estimation, OFDM), and deep reinforcement learning for wireless systems, with prior work in Robotic CPS safety and formal verification.

Education

University of Nebraska-Lincoln – PhD

Graduate Research Assistant – School of Computing

Advised by: Dr. Mehmet Can Vuran, Dale M. Jensen Chair Professor | GPA: 3.75/4.0

Dissertation: ML-Driven Beam Management and Power Control for mmWave Vehicular Networks

Nebraska, USA

June 2024 – Present

University of Nebraska-Lincoln – Master's

Graduate Research Assistant – School of Computing

Advised by: Dr. Dung Hoang Tran, Assistant Professor | GPA: 3.75/4.0

Nebraska, USA

Aug 2021 – May 2024

Guru Gobind Singh Indraprastha University – Bachelor's

Electronics and Communication

GPA: 7.9/10 | Thesis: Deep learning techniques for MRI denoising and reconstruction from undersampled MRI scans.

Delhi, India

Aug 2017 – May 2021

Research Experience

University of Nebraska-Lincoln

Doctor of Philosophy, Computer Science

- **Camera-Primed Double-Directional Beam Management (IEEE SECON 2026, accepted):** Developed a vision-assisted mmWave beamforming framework enabling real-time TX/RX beam selection using camera-derived spatial cues and adaptive correction under hardware constraints.
- **Reinforcement Learning for mmWave Power Control:** Designed a hardware-in-the-loop RL framework to optimize transmit power and RF gain parameters under SNR constraints using SDR-based testbeds and real-time feedback.
- Utilized ray-tracing tools (**Wireless InSite**, **Sionna RT**) to model propagation environments and validate experimental observations.
- **Technical Skills:** GNU Radio, USRP, Python, C++, Ubuntu Linux, Git

Nebraska, USA

Aug 2024 – Current

University of Nebraska-Lincoln

Master's, Computer Science

- Perception-based Runtime Monitoring and Verification for Human-Robot Construction Systems
- Experience with **Neural Network Verification (NNV)** software for formal analysis of deep learning models.
- Knowledge of **Signal Temporal Logic (STL)** for specifying and verifying temporal properties in cyber-physical systems.
- **Technical Skills:** ROS, Gazebo, RVIZ, HTML, CSS, Python, C++, Ubuntu Linux, Git.

Nebraska, USA

Aug 2021 – May 2024

Publications

CONFERENCE PROCEEDINGS

Look Once, Beam Twice: Camera-Primed Real-Time Double-Directional mmWave Beam Management for Vehicular Connectivity

Avhishek Biswas, Apala Pramanik, Eylem Ekici, Mehmet Can Vuran

2026 IEEE International Conference on Sensing, Communication, and Networking (SECON), 2026

Perception-based Runtime Monitoring and Verification for Human-Robot Construction Systems

Apala Pramanik, Sung Woo Choi, Yuntao Li, Luan Viet Nguyen, Kyungki Kim, Hoang-Dung Tran

2024 22nd ACM-IEEE International Symposium on Formal Methods and Models for System Design (MEMOCODE), 2024

ASTITVA: assistive special tools and technologies for inclusion of visually challenged

Apala Pramanik, Rahul Johari, Nitesh Kumar Gaurav, Sapna Chaudhary, Rohan Tripathi

2021 international conference on computing, communication, and intelligent systems (ICCCIS), 2021

START: Smart Stick based on TLC Algorithm in IoT Network for Visually Challenged Persons

Rahul Johari, Nitesh Kumar Gaurav, Sapna Chaudhary, Apala Pramanik

2020 Fourth International Conference on I-SMAC (IoT in Social, Mobile, Analytics and Cloud)(I-SMAC), 2020

WORKSHOPS

Vision-based Runtime Monitoring for Human-Construction Robot Systems

Apala Pramanik, Kyungki Kim, Dung Hoang Tran

IROS 2023 Workshop on Formal Methods Techniques in Robotics Systems: Design and Control, 2023

University Projects

mmWave Propagation Engineering and Bistatic Radar Characterization
University of Nebraska–Lincoln

Nebraska, USA
Spring 2025

- Investigated mmWave propagation mechanisms by leveraging near-critical incidence on dielectric surfaces to induce lateral wave components for extended coverage along walls.
- Designed controlled experiments to characterize angle-dependent reflection and surface-guided propagation behavior in indoor environments.
- Developed a bistatic radar measurement framework using liquid metal (EGaIn) and metallic reflectors to study material-dependent scattering and signal enhancement effects.
- Technical Skills:** mmWave Propagation, Bistatic Radar, RF Measurement, USRP/SDR, Signal Analysis, Experimental Design
- Soft Skills:** Research Methodology, Problem Formulation, Data Interpretation, Technical Communication

Multi-Modal Traffic Pattern Detection Using Radio and Camera Sensing
University of Nebraska–Lincoln

Nebraska, USA
Fall 2024

- Built a real-time multi-modal traffic classification system using synchronized USRP RF signals and camera images for three-class detection.
- Implemented real- and complex-valued CNN models, achieving >80% accuracy with <100 ms inference latency.
- Integrated precise timestamp synchronization (± 10 ms) and a watchdog mechanism for reliable, continuous operation.
- Technical Skills:** Python, PyTorch/TensorFlow, USRP, Signal Processing, Deep Learning (CNN, CVNN), Multi-Modal Data Fusion
- Soft Skills:** System Design, Cross-Disciplinary Collaboration, Technical Presentation, Analytical Problem Solving

mmWave 3D Ray Tracing Framework
University of Nebraska–Lincoln

Nebraska, USA
Fall 2024

- Built a MATLAB-based 3D ray tracing simulator for mmWave indoor propagation with multi-bounce reflections and log-normal shadowing.
- Implemented log-distance path loss modeling and reflection loss computation at 24 GHz.
- Developed modular environment modeling and power-gradient visualization with automated SVG/EPS export and Python-based GIF generation.
- Technical Skills:** MATLAB, Python, Ray Tracing, Wireless Channel Modeling

Deep Learning and NLP Projects
University of Nebraska–Lincoln

Nebraska, USA
Aug 2021 – July 2024

- Developed a Spanish news classification system using GRU, LSTM, and BETO models, achieving 91.80% accuracy with BETO and tokenizer optimization.
- Applied data augmentation techniques (synonym and character replacement) to expand the dataset, improving training accuracy to 99%.
- Implemented image classification models on Fashion-MNIST and CIFAR-100 and performed sentiment analysis using sequential deep learning architectures.
- Technical Skills:** Python, Pandas, Matplotlib, Seaborn, TensorFlow, Deep Learning (CNNs, RNNs, Transformers)
- Soft Skills:** Analytical Thinking, Technical Writing, Teamwork, Presentation Skills

Involvement

Peer Reviewer	IEEE Transactions on Wireless Communications (2025, 2026)
Undergrad Research Fair	Presented research projects to undergraduate computer science major students
K-12 Outreach Fair	Demonstrated research activities to K–12 students
Mentorship	Mentored undergraduate researchers on experimental design, data analysis, and research communication

Skills

ML / DL	PyTorch, TensorFlow, Scikit-learn, NumPy, Pandas
Wireless / SDR	GNU Radio, USRP, Wireless InSite, Sionna RT, mmWave, 5G/6G, V2X/V2I, Beamforming, OFDM, Channel Estimation
Languages	Python, C++, MATLAB, Bash/Zsh
Robotics / CPS	ROS (rospy, roscpp), Gazebo, RVIZ, Point Cloud Library
Tools	Git, Linux, L ^A T _E X, Arduino, Proteus
Coursework	Deep Learning (A+), Wireless Comm Networks (A+), Advanced Wireless Comm (A), Hardware Acceleration for ML (B+), IoT (A+)
Soft Skills	Mentoring, Time Management, Teamwork, Problem-solving, Documentation, Engaging Presentation

Interests

Paper Review	I enjoy critically reviewing research papers to evaluate technical rigor, clarity of methodology, and the strength of scientific contributions.
Technical Writing	I enjoy writing research papers to convey my research work in easy and understandable language.
Dancing	I have a senior diploma degree in Bharatnatyam (Indian Classical Dance) and have training of more than 10 years.
Fitness	I am passionate about weight training and maintaining a healthy and active lifestyle.

References available upon request.